# Offshore–nearshore wave dependence

An information-theoretic comparison of two NSW coastal settings

- Figure Eight Pools and Stockton
- MSc Information Theory project
- Descriptive deployment comparisons • MI and CMI

# Why compare these settings?

Physical motivation and predictions

- Kinsela et al. (2024): greater height scatter at embayed Stockton
- Height MI predicted to be greater at exposed Figure Eight Pools
- Period and direction CMI predicted to be greater at Stockton
- Question: what differences do the observed deployment records show?

# Two buoy pairings, different deployments

Sampling intervals and complete concurrent observations

- Port Kembla / Wollongong → Figure Eight Pools: N = 6,867
- Sydney → Stockton: N = 10,522
- Offshore: hourly parameters • Nearshore: half-hourly parameters
- Matched: effectively hourly, with missing-data gaps
- FEP: Dec 2016–Dec 2017 • Stockton: Dec 2019–Apr 2021

# From buoy records to aligned observations

Quality control without filling missing hours

- Ho = offshore WHTH; Hn = nearshore Hsig: time-domain significant height
- Tp,o = offshore WPPE; direction = offshore WPDI relative to true north
- Apply supplied quality flags; align exact UTC timestamps
- Retain valid extreme waves; no interpolation
- Direction is one joint predictor: (cos θ, sin θ)

# What do MI and CMI measure?

Dependence beyond a linear correlation

- Height MI: I(Ho; Hn)
- Period CMI: I(Tp,o; Hn | Ho)
- Direction CMI: I((Dx, Dy); Hn | Ho)
- CMI asks what remains associated with Hn after Ho is known
- Bits are not variance explained, attenuation or a causal effect

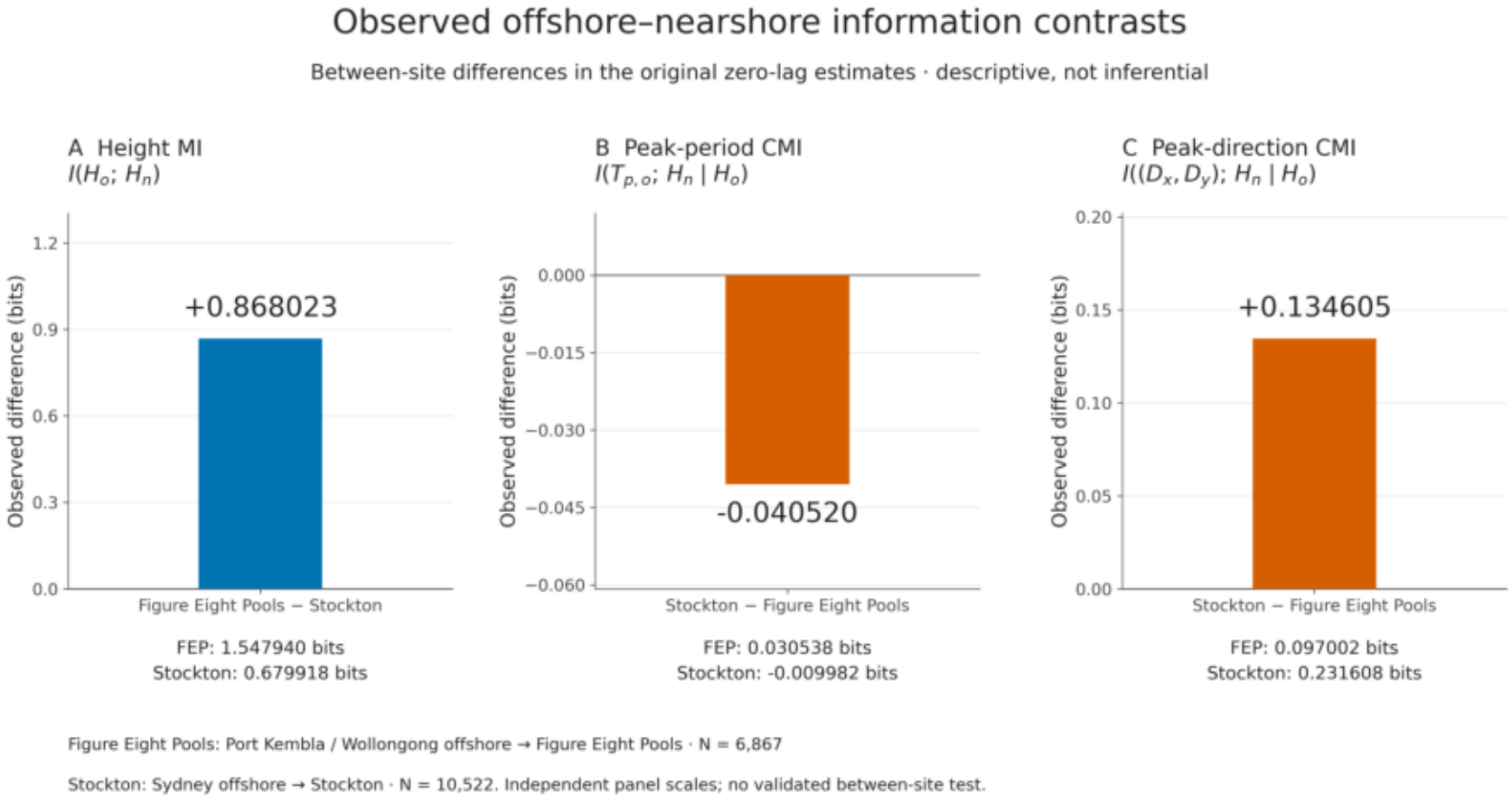
# A consistent estimator across both sites

Verified handling of temporal gaps

- JIDT continuous Kraskov algorithm 1 • k = 4
- Normalization • max norm • GPU disabled
- 45-hour Theiler exclusion using actual elapsed time
- Across-gap Java neighbour checks: 44 / 45 h excluded; 46 h eligible
- Tie noise: 1e−8 after normalization • fixed seed 20261031

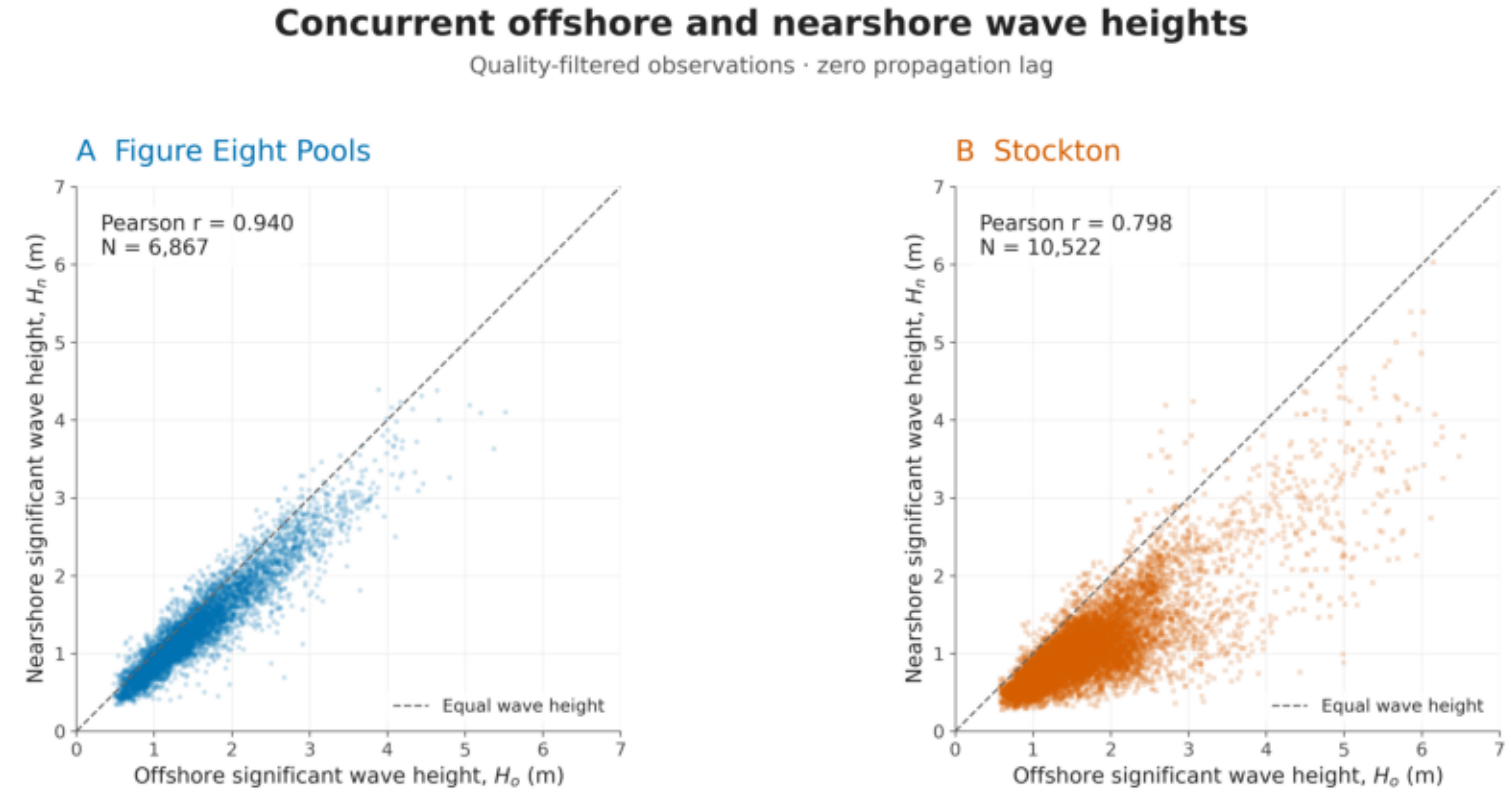
# Main results: direct site comparisons

Original zero-lag estimates • bits • descriptive only



# Height relationships in the observations

All complete concurrent samples • matching axes

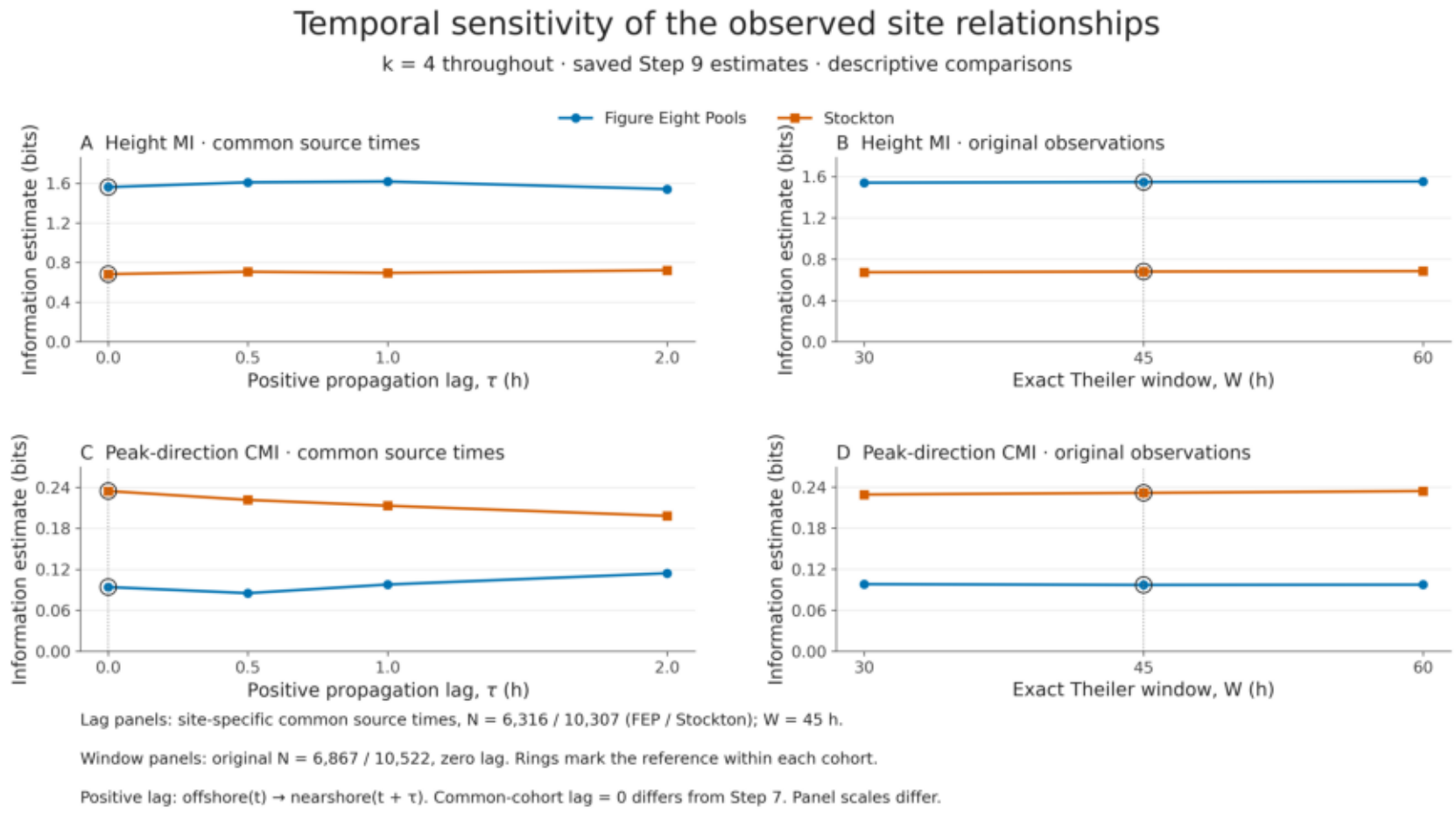



Matched time-domain significant wave heights (offshore WHTH; nearshore Hsig). Identical axes and equal aspect in both panels.

Pearson correlations describe linear association; the dashed line is a reference, not a fitted model.

# Do the site orderings persist?

Propagation lag and Theiler-window sensitivity • k = 4



# Physical interpretation: partial support

Consistent patterns, without a causal claim

- Height prediction: consistent with the observed estimates
- Direction prediction: consistent; contrast magnitude depends on lag
- Period prediction: not supported by the observed ordering
- Exposure, sheltering and refraction are plausible explanations
- Period may overlap with height information; mechanisms were not tested

# What the comparison can—and cannot—show

Statistical and environmental limitations

- One candidate between-site interval method failed initial calibration
- No validated wave-data confidence intervals or p-values
- Autocorrelation, unequal N, gaps and different seasonal coverage
- Different depths and offshore stations; one site per setting
- Sensitivity supports descriptive ordering, not population significance

# Conclusion

Two deployment records show contrasting information relationships

- Height MI: greater at Figure Eight Pools by 0.868023 bits
- Direction CMI: greater at Stockton by 0.134605 bits
- Period CMI: small; Stockton − FEP = −0.040520 bits
- Height and direction orderings persist under temporal checks
- Descriptive contrasts—not significant population or causal effects

# References and reproducibility

Supporting slide—not part of the timed narration


- Kinsela et al. (2024), Scientific Data 11, 190
- doi:10.1038/s41597-023-02865-x
- Kraskov et al. (2004), Physical Review E 69, 066138
- Lizier (2014), Frontiers in Robotics and AI 1, 11
- Sources: final report, primary-information CSV and saved sensitivity results